



ASSIGNMENT REPORT

Digital Design

Asynchronous FIFO Depth Calculation

Course	Digital Design
Assignment	FIFO Depth Calculation – Examples 1, 2 & 3
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01 OBJECTIVE

This assignment determines the minimum required depth of an asynchronous FIFO that transfers a fixed-size burst of data between two modules running on independent clock domains, for three scenarios that differ in the idle (wait) cycles present on the write and read sides.

02 PROBLEM STATEMENT / REQUIREMENTS

Three worked examples are analyzed for an asynchronous FIFO transferring data from a faster writing module (A) to a slower reading module (B). In each case the FIFO must be sized so that it never overflows during the burst. The common and per-example parameters are:

- Writing clock $f_A = 100$ MHz ($T_A = 10$ ns); Reading clock $f_B = 40$ MHz ($T_B = 25$ ns); Burst Length (BL) = 100 data items — common to all three examples.
- Example 1: no idle cycles on either the write side or the read side (back-to-back writes and reads).
- Example 2: 2 idle cycles between successive writes and 2 idle cycles between successive reads.
- Example 3: 2 idle cycles between successive writes and 1 idle cycle between successive reads.

03 DESIGN & ANALYSIS

3.1 Design Approach

Because $f_A > f_B$ in every example, module A always writes faster than module B can read, so FIFO occupancy grows throughout the burst and peaks when the last item is written. The minimum depth is found in three steps: (1) find the effective write period $T_{A'}$ and read period $T_{B'}$ by adding any idle cycles to the base period; (2) find the total time needed to write the whole burst, $T_W = BL \times T_{A'}$; (3) find how many items B has managed to read during that same time, $N_{read} = T_W / T_{B'}$. The minimum required FIFO depth is then $Depth = BL - N_{read}$.

3.2 Given Data & Calculation

The table below lists the parameters and intermediate results for each example, derived using the method described in 3.1.

Parameter	Example 1	Example 2	Example 3
Write clock, f_A	100 MHz	100 MHz	100 MHz
Read clock, f_B	40 MHz	40 MHz	40 MHz
Write period, T_A	10 ns	10 ns	10 ns
Read period, T_B	25 ns	25 ns	25 ns
Idle cycles / write	0	2	2
Idle cycles / read	0	2	1
Effective write period, $T_{A'}$	10 ns	30 ns	30 ns
Effective read period, $T_{B'}$	25 ns	75 ns	50 ns
Burst Length, BL	100	100	100
Total write time, $T_W = BL \times T_{A'}$	1000 ns	3000 ns	3000 ns
Items read, $N_{read} = T_W / T_{B'}$	40	40	60
Minimum FIFO depth = $BL - N_{read}$	60	60	40

Table 1: Calculation parameters and results for Examples 1, 2 and 3.

3.3 Async FIFO Architecture

All three examples share the same generic architecture: module A writes into the FIFO on its own clock domain, and module B reads from the FIFO on an independent, slower clock domain, as shown below.

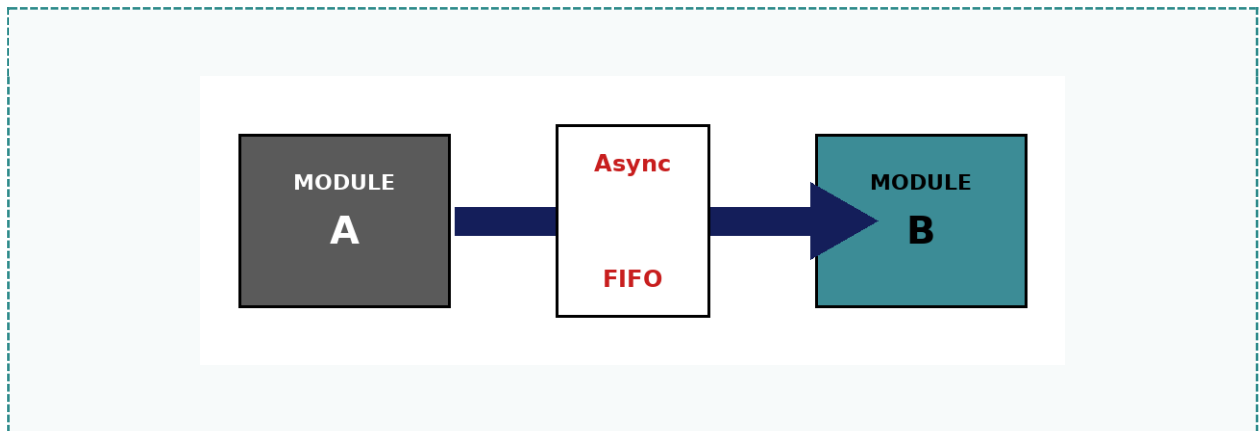


Figure 1: Generic asynchronous FIFO connecting Module A (write clock domain) and Module B (read clock domain).

3.4 Step-by-Step Calculation

The arithmetic performed for each example, following the method in 3.1, is shown below.

Example 1 (0 idle cycles write, 0 idle cycles read):

$$T_W = BL \times T_A = 100 \times 10 \text{ ns} = 1000 \text{ ns}$$

$$N_{\text{read}} = T_W / T_B = 1000 / 25 = 40 \text{ items}$$

$$\text{Depth} = BL - N_{\text{read}} = 100 - 40 = 60 \text{ items}$$

Example 2 (2 idle cycles write, 2 idle cycles read):

$$T_{A'} = 3 \times T_A = 30 \text{ ns} \quad T_{B'} = 3 \times T_B = 75 \text{ ns}$$

$$T_W = BL \times T_{A'} = 100 \times 30 \text{ ns} = 3000 \text{ ns}$$

$$N_{\text{read}} = T_W / T_{B'} = 3000 / 75 = 40 \text{ items}$$

$$\text{Depth} = BL - N_{\text{read}} = 100 - 40 = 60 \text{ items}$$

Example 3 (2 idle cycles write, 1 idle cycle read):

$$T_{A'} = 3 \times T_A = 30 \text{ ns} \quad T_{B'} = 2 \times T_B = 50 \text{ ns}$$

$$T_W = BL \times T_{A'} = 100 \times 30 \text{ ns} = 3000 \text{ ns}$$

$$N_{\text{read}} = T_W / T_{B'} = 3000 / 50 = 60 \text{ items}$$

$$\text{Depth} = BL - N_{\text{read}} = 100 - 60 = 40 \text{ items}$$

04 RESULTS SUMMARY

The table below summarizes the minimum FIFO depth obtained for each example.

Example	Idle Cycles (Write / Read)	Items Read During Burst	Minimum FIFO Depth
1	0 / 0	40	60
2	2 / 2	40	60
3	2 / 1	60	40

05 DISCUSSION

Examples 1 and 2 give the same minimum depth (60 items) even though idle cycles are introduced in Example 2, because the ratio between the effective read and write periods is unchanged ($T_B'/T_A' = 2.5$ in both cases) — adding equal proportional idle time on both sides does not change the relative speed difference between A and B. Example 3 requires a smaller depth (40 items) because the read side has fewer idle cycles than the write side, so B reads relatively faster compared to A than in Examples 1 and 2, and fewer items accumulate in the FIFO before B catches up. In a real design, these minimum depths would typically be rounded up to the next convenient binary size (e.g. 64) since FIFO read/write pointers are usually implemented with binary or Gray-code counters.

06 CONCLUSION

Using a burst-time / read-time analysis, the minimum FIFO depth needed to avoid overflow was found to be 60 items for Examples 1 and 2, and 40 items for Example 3. The key takeaway is that the required FIFO depth depends on the relative effective throughput of the write and read clock domains — including any idle cycles on either side — rather than on the raw clock frequencies alone.

A APPENDIX – HANDWRITTEN CALCULATIONS

Scanned original hand-written calculations for Examples 1, 2 and 3, corresponding to the results presented in Sections 3.2–3.4 above.

